USN | | | | | | | | | | | [23BSBE01]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

BIOLOGY FOR ENGINEERS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.
ii. Answer all questions.

		PART - A	**05 X 02=10**		
1.	**a.**	State cell theory.	L-2	CO1	**02**
	b.	Classify the bacteria based on their shape. Give an example for each.	L-2	CO2	**02**
	c.	Define enzyme specificity.	L-2	CO3	**02**
	d.	Explain glycolysis process.	L-1	CO4	**02**
	e.	Describe sterilization? Write its three types.	L-2	CO5	**02**
		PART - B	**06 X 05=30**		
2.	**a.**	Write the salient features of DNA.	L-1	CO1	**05**
		OR			
	b.	Briefly discuss three types of RNAs and their biological roles.	L-2	CO1	**05**
3.	**a.**	What does kingdom Protista consist of? Write its five classifications.	L-1	CO2	**05**
		OR			
	b.	Illustrate the notable features of Kingdom fungi. Mention their classifications.	L-3	CO2	**05**
4.	**a.**	Explain the Lock and Key hypothesis of an enzyme action with neat labelled diagram.	L-3	CO3	**05**
		OR			
	b.	Describe the enzymes involved in the digestion process.	L-2	CO3	**05**
5.	**a.**	Explain exergonic reactions with a neat energy profile diagram.	L-2	CO4	**05**
		OR			
	b.	Describe the cellular respiration process.	L-2	CO4	**05**
6.	**a.**	What is media composition? Discuss the essential components of culture media.	L-1	CO5	**05**
		OR			
	b.	Elaborate any five methods used for the identification of microorganisms.	L-3	CO5	**05**
7.	**a.**	Discuss the components of prokaryotic cell with neat diagram.	L-2	CO1	**05**
		OR			
	b.	Write a note on ecological aspects of single cellular organisms.	L-2	CO5	**05**
		PART - C	**06 X 10=60**		
8.	**a.**	Discuss the structure and components of Eukaryotic cell in detail with a neat diagram.	L-2	CO1	**10**

		OR			
	b.	**i.** Define pharmacokinetics and its four phases (ADME). **ii.** Describe vaccines. Discuss the three important types of vaccines.	L-2	CO1	**10**
9.	**a.**	Write the characteristic features of kingdom animalia and plantae?	L-1	CO2	**10**
		OR			
	b.	Classify the organisms based on energy utilization.	L-2	CO2	**10**
10.	**a.**	Explain the DNA replication process with the detailed role of each enzyme involved.	L-2	CO3	**10**
		OR			
	b.	Narrate the five stages involved in enzyme catalyzed reactions with a neat energy profile diagram?	L-3	CO3	**10**
11.	**a.**	Write the salient features of energy yielding and energy consuming reactions with suitable examples.	L-1	CO4	**10**
		OR			
	b.	Draw the chemical structure of ATP and explain how it is referred as energy currency.	L-3	CO4	**10**
12.	**a.**	**i.** Illustrate the biological species concept and its limitations **ii.** Classify microorganism based on taxonomy, morphology, physiology, and ecological roles.	L-3	CO5	**10**
		OR			
	b.	**i.** Discuss the salient features of viruses **ii.** Describe microscope and write its three types.	L-2	CO5	**10**
13.	**a.**	**i.** Write the various types involved in protein structure with neat diagrams. **ii.** Write a short note on central dogma of life.	L-1	CO1	**10**
		OR			
	b.	**i.** Define chemotherapy and discuss its four types. **ii.** What is half-life of a drug. Write the formulas for its calculation.	L-2	CO1	**10**